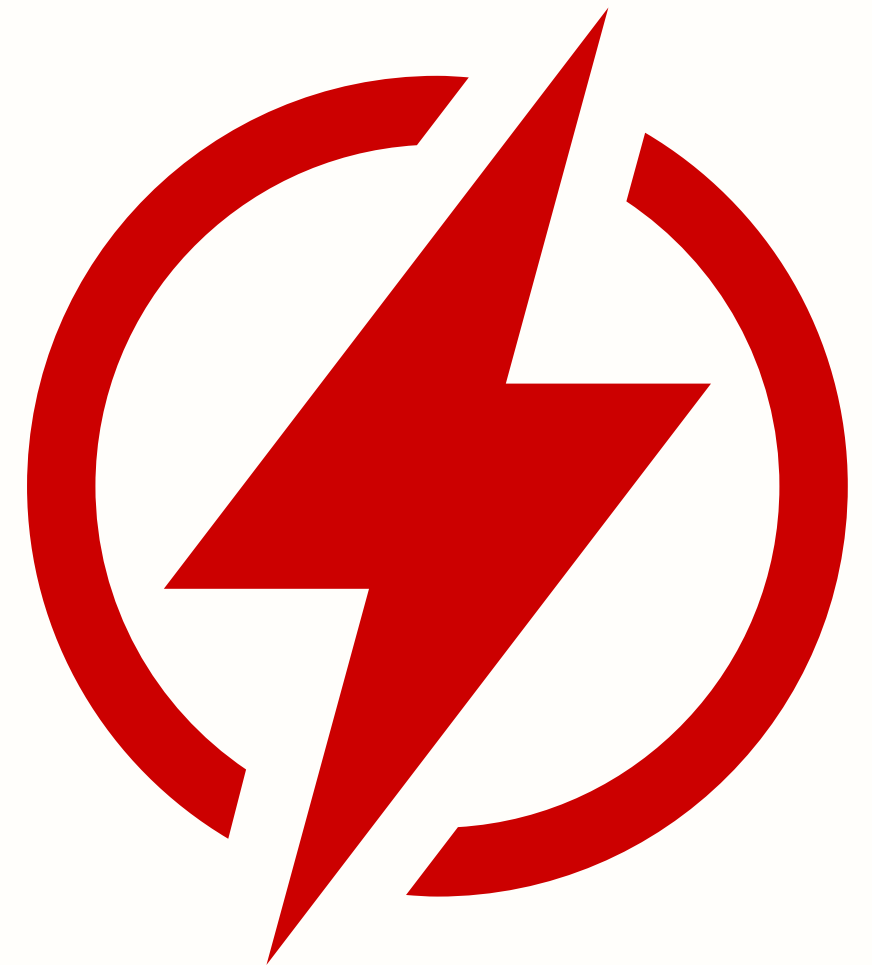


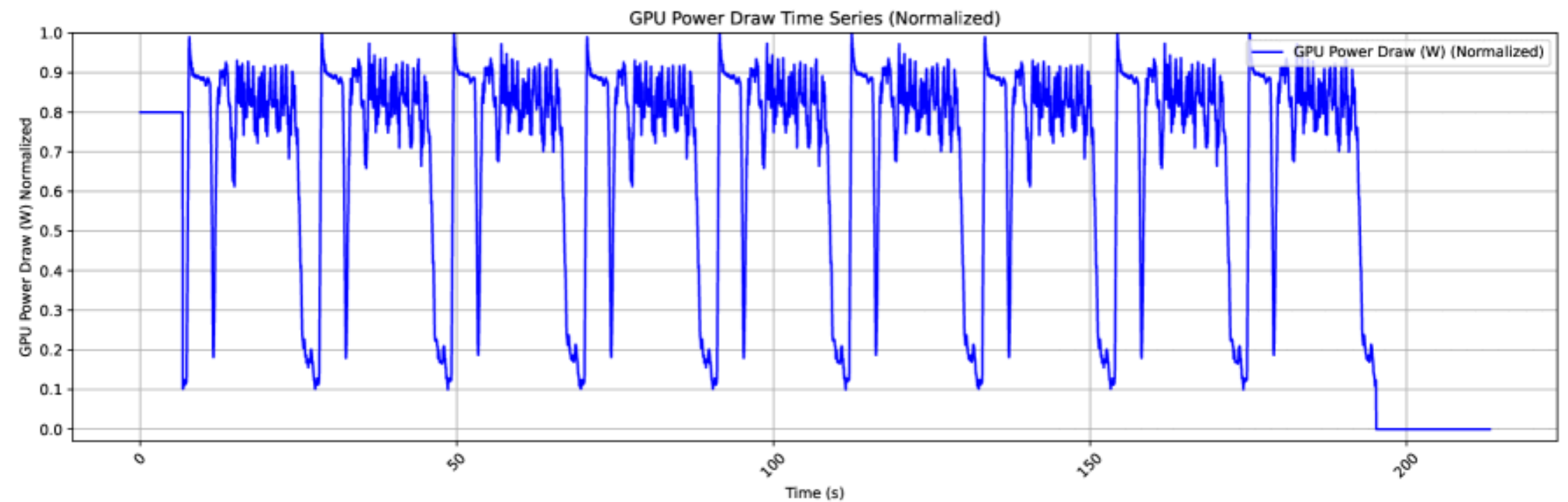
COLLECT & ADJUST SERVER LEVEL POWER CONSUMPTION

Emily, Alex, Daniel

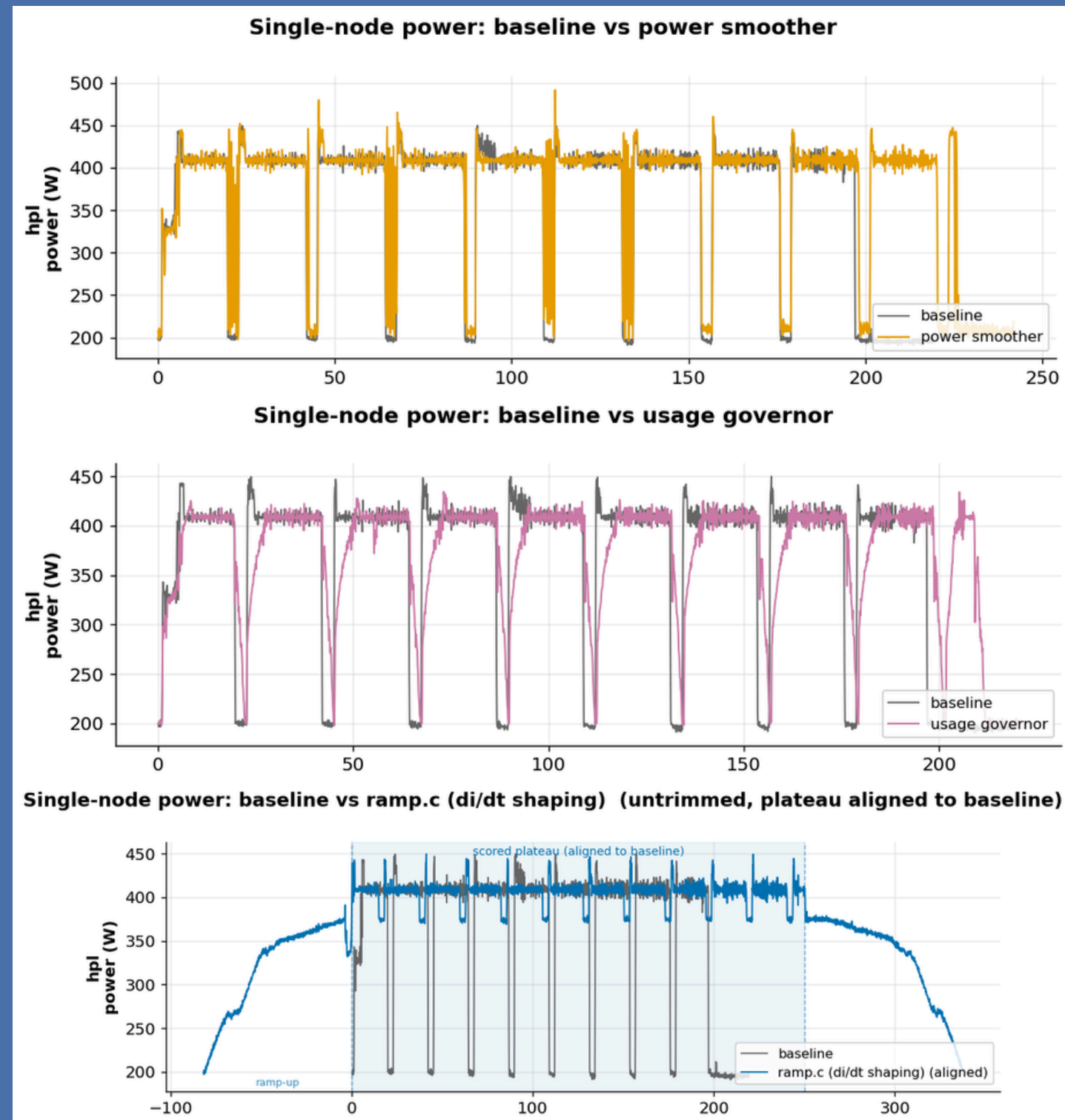


THE PROBLEM

- HPC jobs → Power fluctuations
 - Synchronous (AI training)
- Grid impacts
 - damage to hardware, etc



OUR SOLUTIONS



- 3 distinct mechanisms
 - “smoothers”
- **Goal:** Manage abrupt power swings
 - Rise → “spikes”
 - Limit increase
 - Ramp up
 - Fall → “drops”
 - Burn power
 - Ramp down

RAPL & REDFISH

- For server power:
 - BMC Redfish
 - RAPL
 - High freq polling
- Power capping

Table 2. Telemetry Comparison

	RAPL	Redfish
Scope	CPU package, cores, DRAM	Whole chassis
Granularity	Per-socket, per-domain	Chassis total
Useful rate	~10 Hz	~1 Hz
Path	In-band, host CPU	Out-of-band, BMC
Host overhead	<3% of one core	None
Requires	Root on host	BMC credentials + network
Survives host hang	No	Yes
Capping	Yes — millisecond-window PL1/PL2	Yes — coarse, slow (DCMI)
Portability	x86, vendor-specific	Vendor-neutral standard

MONITORING POWER & HARDWARE HEALTH

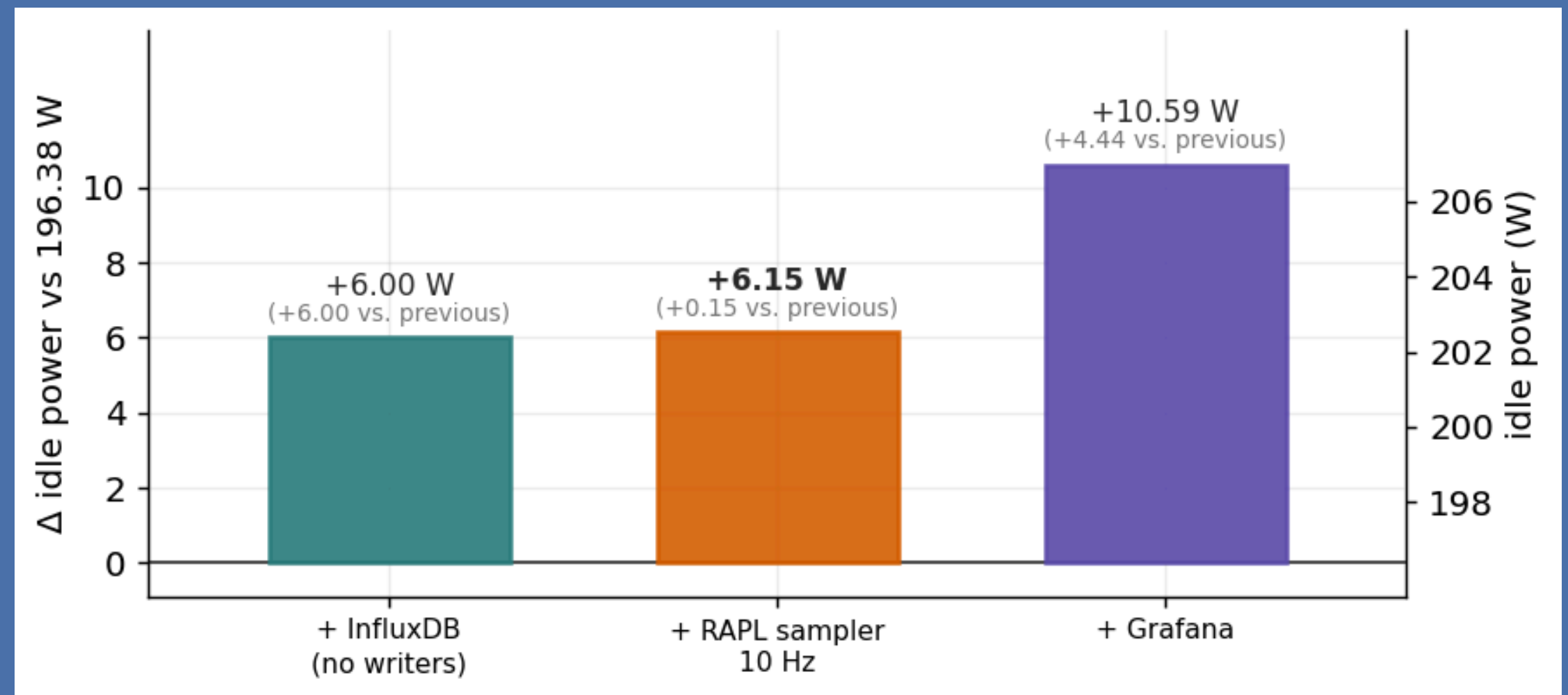
- Collect data from
 - Linux system telemetry
 - e.g. /proc/stat
 - PDU
 - Redfish
- Daemons

LIVE DASHBOARD



THE STACK

- Live graphs
- Data → InfluxDB → Grafana
- Background daemons
 - RAPL
 - BMC Redfish
 - PDU
 - Linux system telemetry



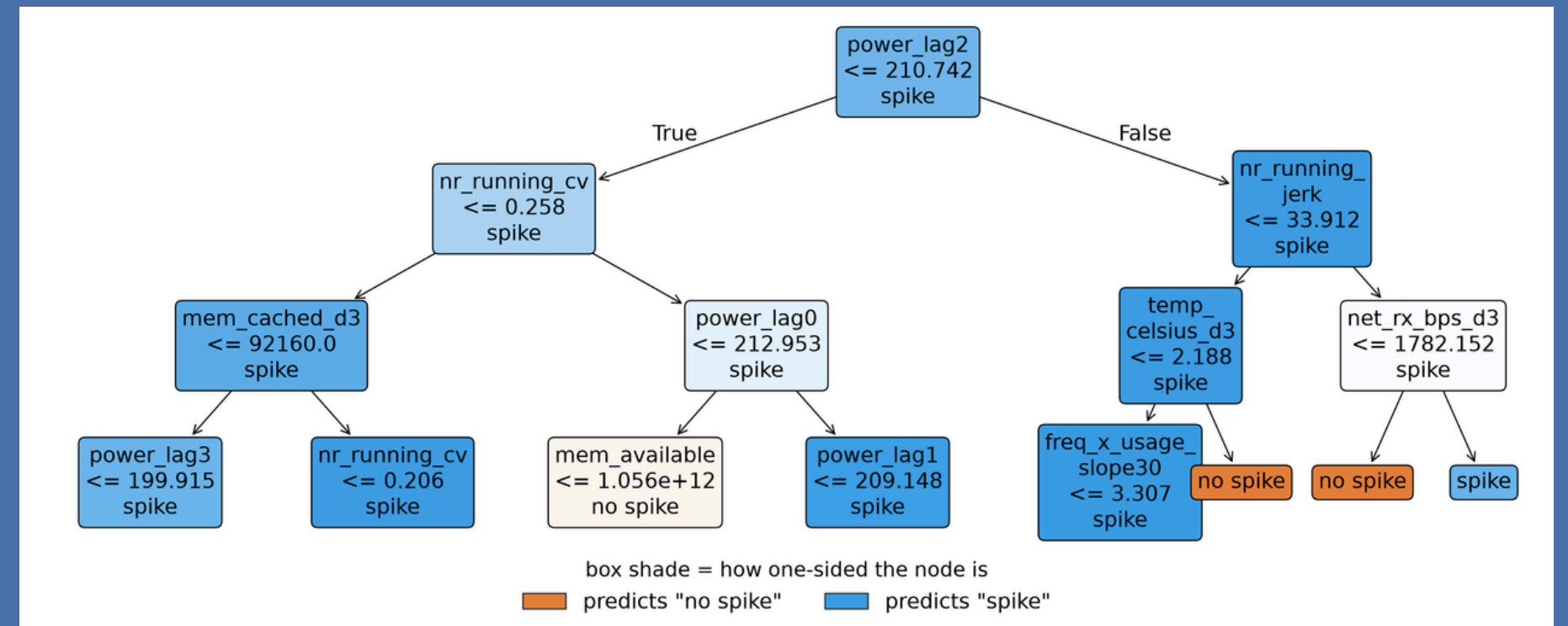
ML/MODEL

- Need right timing for swings
 - Secondary workloads
- Regression (OLS) → poor
- Find heuristics to spikes
 - p-value → Significance
 - Cohen's d → Effect Size

Table 7. Candidate precursors: significance vs. effect size

Candidate feature	What it measures	Significant (p < 0.05)?	Cohen's d	Effect	Kept?
Rolling power coefficient of variation (σ/μ)	Relative volatility over a trailing 30 s window	Yes	1.1–1.2	Large	Yes
Rolling power peak-to-mean ratio (max/ μ)	Burstiness over the same window	Yes	1.1–1.2	Large	Yes
Power EWMA z-score	Deviation from a smoothed running mean	Yes	0.09–0.15	Small	No
Server usage slope	Rate of change of CPU utilization	Yes	0.09–0.15	Small	No

- Classification models
- Random Forest + KF
 - RF → unbiased
 - calculate risk, threshold
 - KF cleans signal



- Hardware limit for latency
 - Prediction → Detection
- Changes:
 - stricter spike definition
 - upstream training (queue)
 - branching (Hawkes test)
 - port RF training

- High correlation b/t power, usage
- Inspired usage-based detector
 - “usage_edge”



Table 8. RF vs usage_edge

	RF detector (champion)	usage_edge
Training	Supervised: 47 features (power, KF, memory, thermal, run-queue), refit on a rolling window	None. Two hand-set thresholds, no fit
Trigger	$P(\text{spike within 5 s}) > \text{calibrated threshold (0.29)}$	usage $\geq 20\%$ <i>and</i> (just crossed 20 % <i>or</i> rising $\geq 25\%/s$)
Lag time	10–20 s	0.1 s (sampling floor)
Onsets caught	—	552 / 557
Flag duty	17.7 %	0.34 %
False alarms	~141 alarms/hr	5.2 alarms/hr
On-box overhead	~2 % of a core (inference); 255 % during on-box refit	Negligible

*Both run on the same HPL workload, ai_load.sh

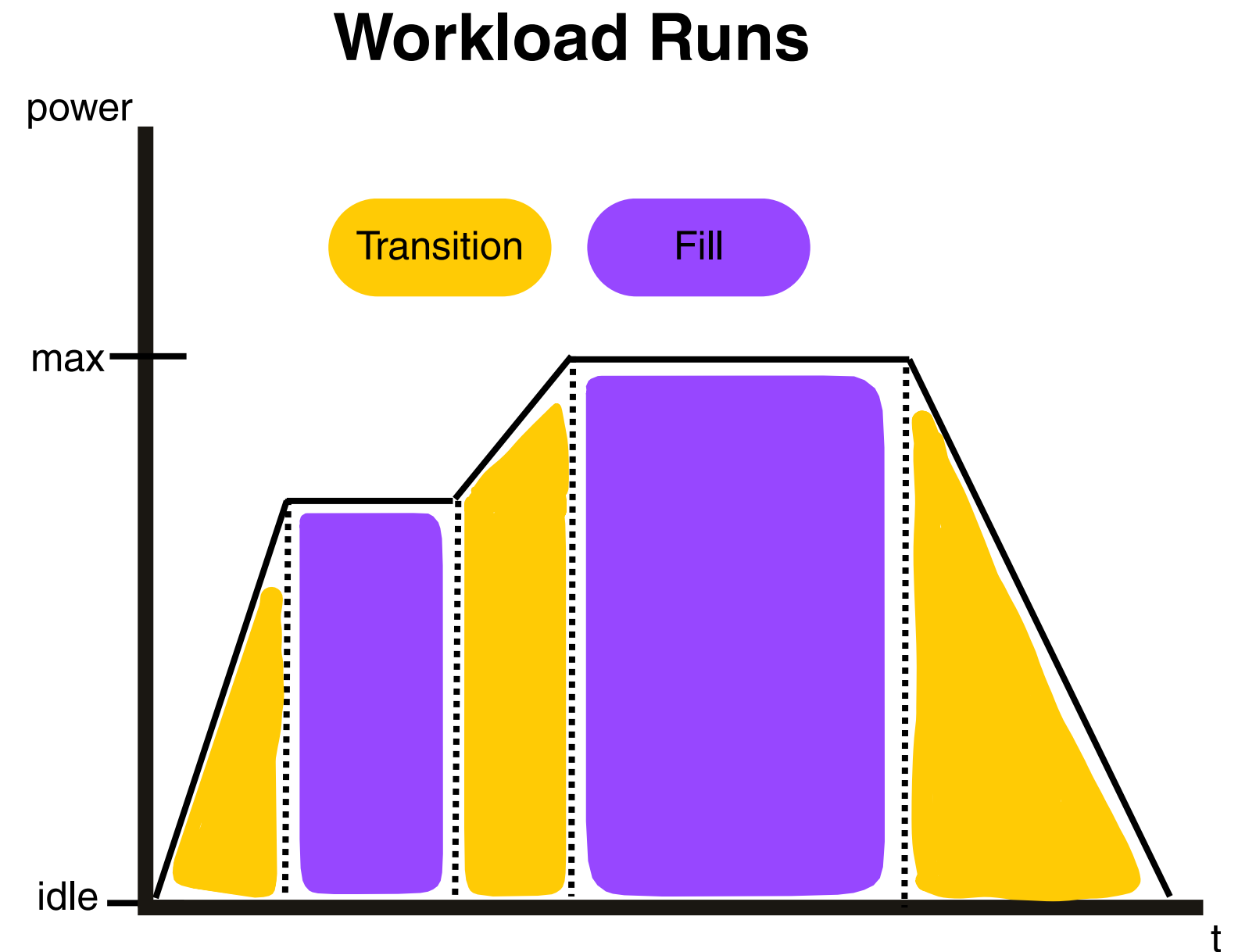
- Beat RF detector considerably
 - overhead, latency, etc
- Retired ML detectors
- Deploy usage_edge in governor

THE SMOOTHERS

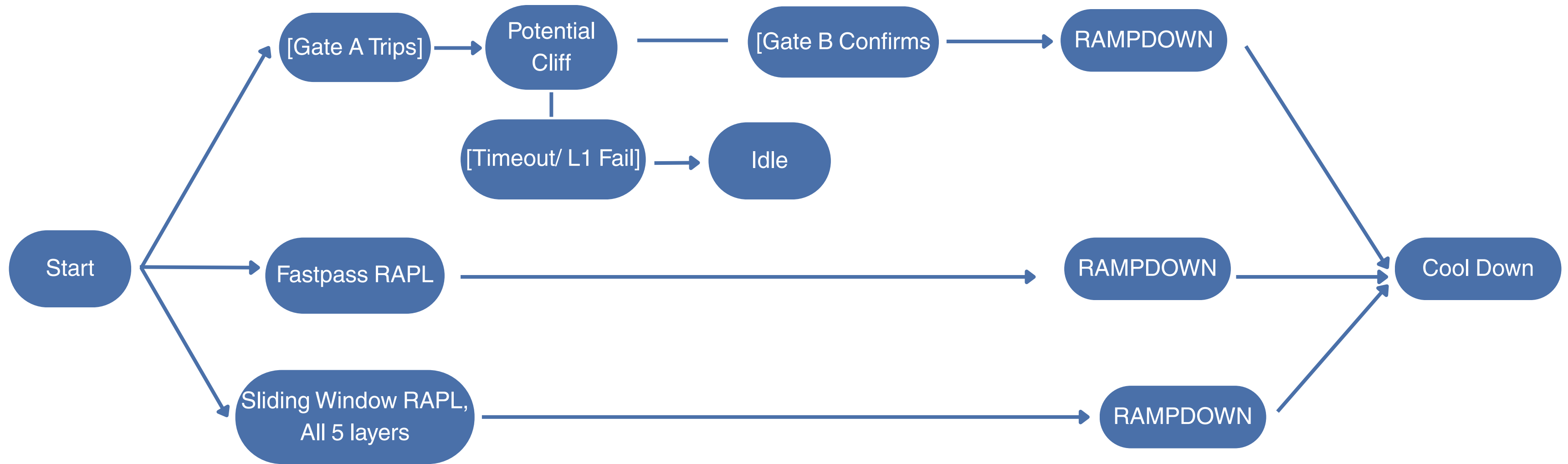
- Ramp.c
 - Trapezoid
- Ramp Down
- Governor
 - Detect swings
 - Ramp up & down

RAMP.C

- Process:
 - Ramp power max
 - Run workload
 - Ramp down
- Manager processes
 - Controls workers
- Transitioning
 - Between primary tasks
- Brute force approach (eats cost!)



RAMP DOWN PROCESS



RAMP DOWN CONT.

Hybrid Gate Detection

- Gate A
 - CPU utilization early tripwire
- Gate B
 - RAPL confirmation

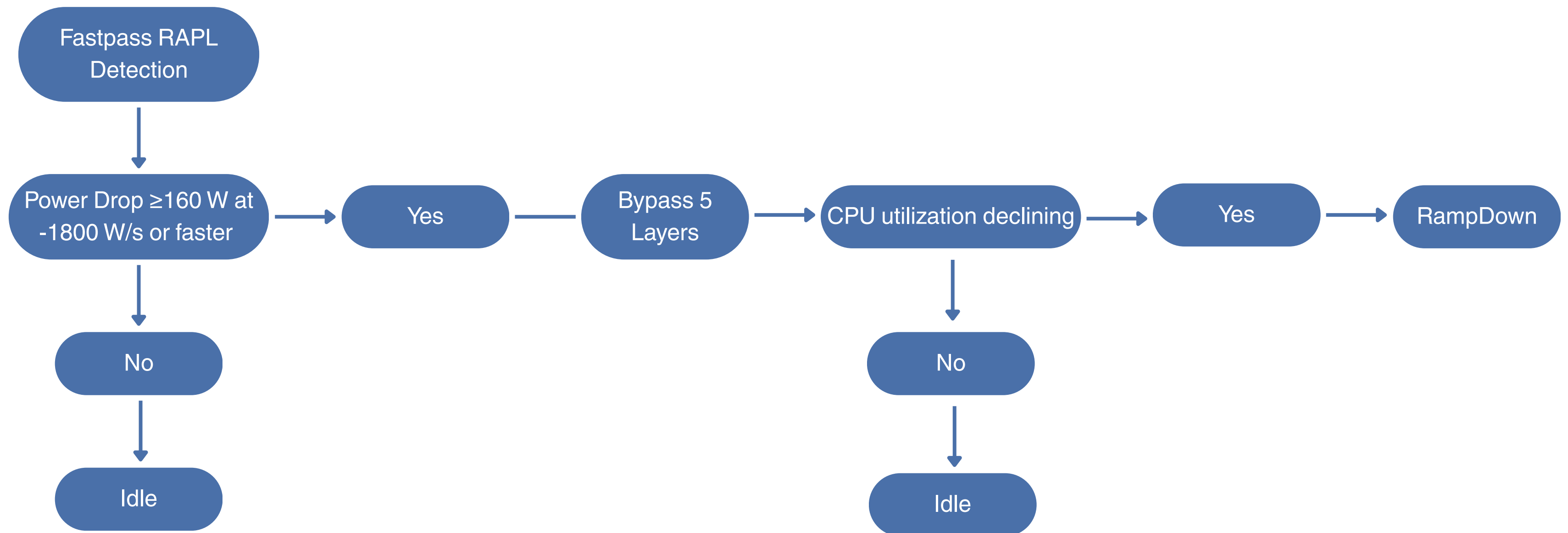
Sliding Window RAPL Path

- Prevent false triggers
- Noise Suppresion

5 Layers

1. Pre-Drop Evaulation Gate
2. Inactive Timeout Gate
3. Rolling Variance Floor
4. Adaptive Pre-Drop History
5. Drop Floor Gate

RAMP DOWN CONT.

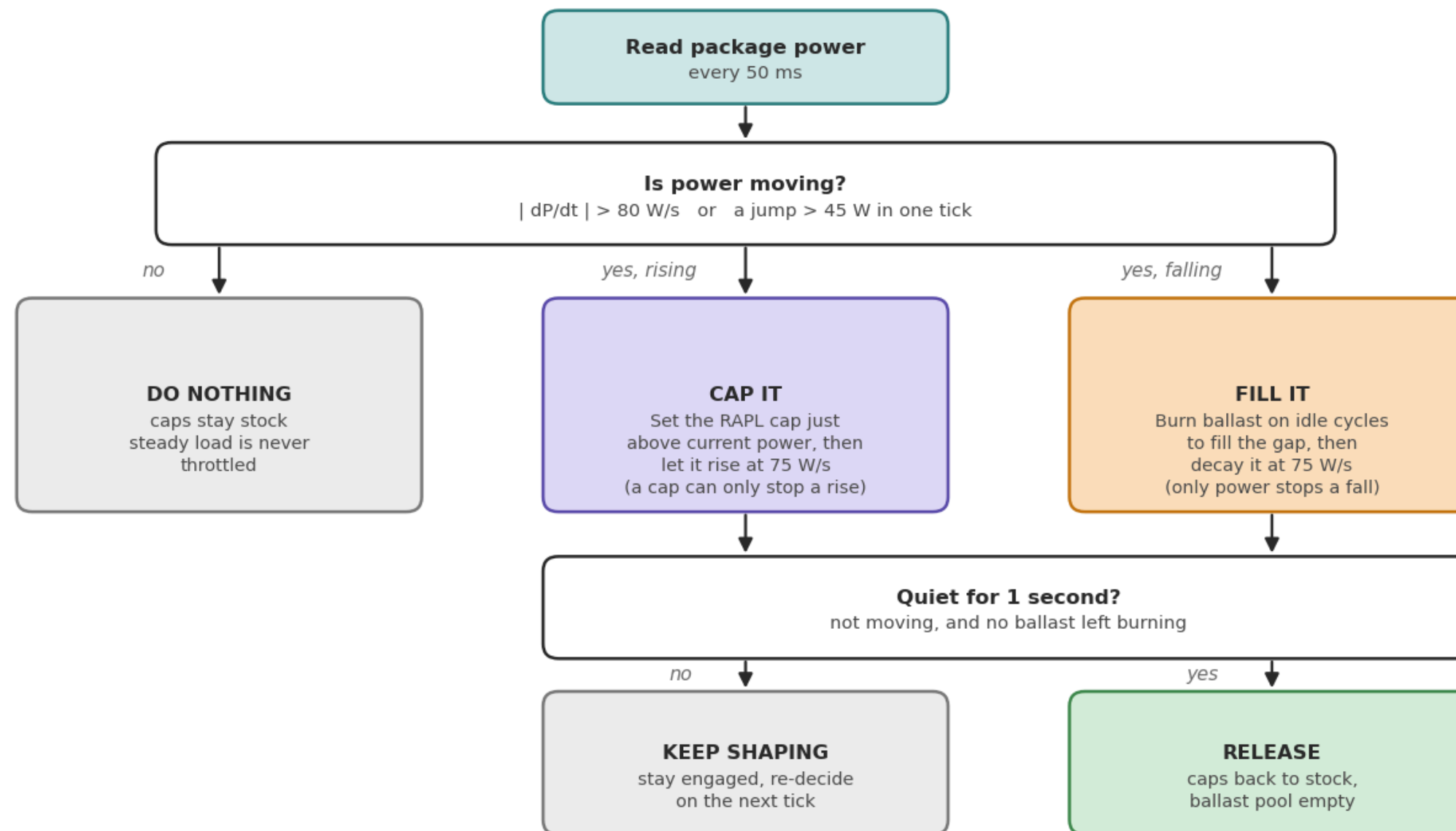


RAMP DOWN CONT.

RampDown

- Control power delivery: 50 ms
 - Max 6 seconds
- Dummy worker intensity
- Abort and Resumption Checks

GOVERNOR



One tick. It repeats 20 times a second.

- usage_edge to detect swings
- Spike → RAPL powercap
- Drop → ballast
 - scheduler assigns duty-cycled jobs from a pool
- Strict swing definition
 - Prone to Type II errors

EXPERIMENT OVERVIEW

- How good are our solutions?
 - 3 Smoothers vs. 3 workloads
 - Mycroft data → server
 - MATLAB → grid
 - Analyze differences

DATA COLLECTION

Run matrix

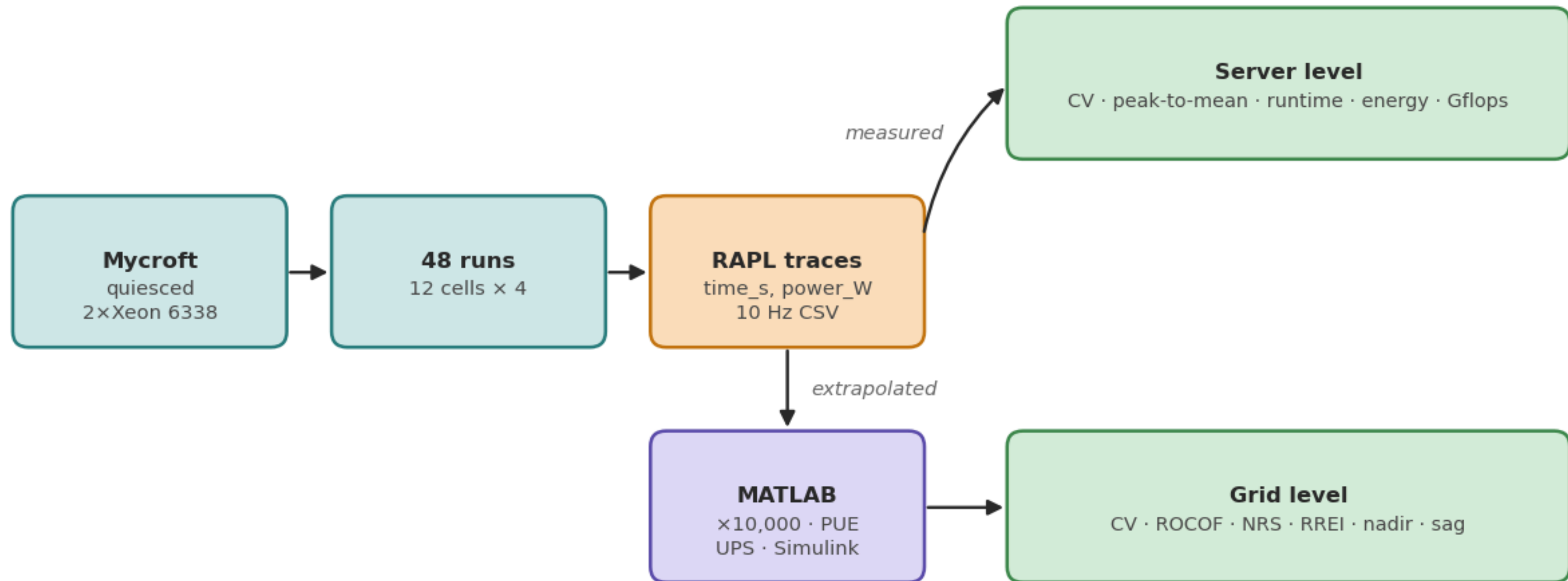
	baseline	power smoother	ramp.c	usagegov
HPL (ai_load.sh)	×4	×4	×4	×4
AI Sim 2	×4	×4	×4	×4
Step	×4	×4	×4	×4

3 workloads × 4 conditions × 4 repeats = 48 traces

Each cell



ANALYSIS PIPELINE



OUR WORKLOADS

Daniel (AI training)

HPL (Linpack)

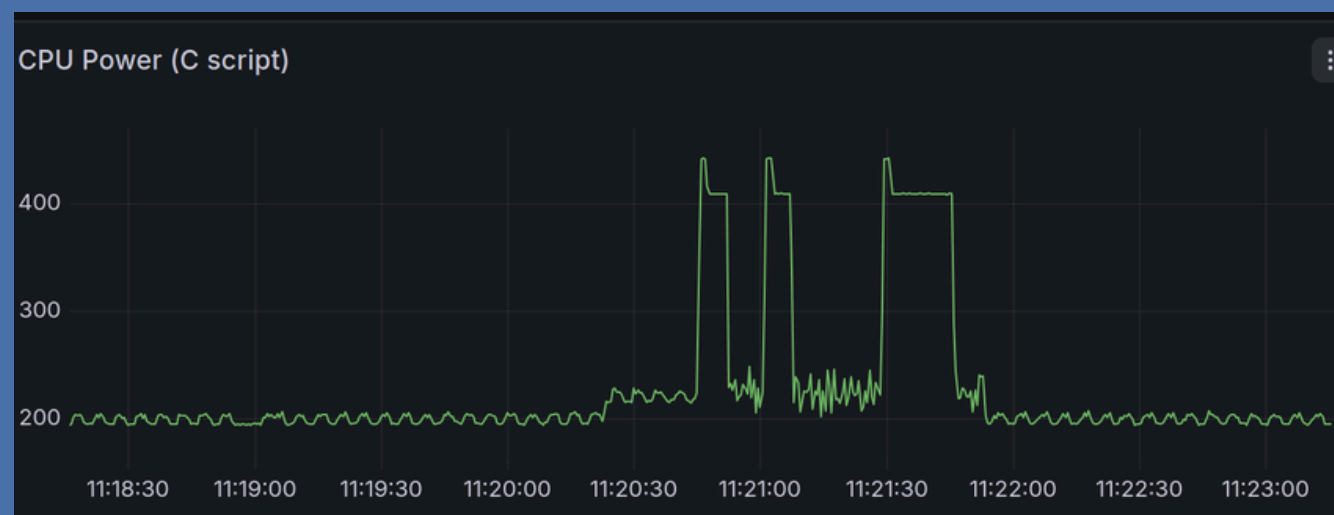
- Completion based
- Mimic AI workload



Emily (random durations)

AI Sim 2

- Randomized
- Mimic AI workload



Alex (customizable steps)

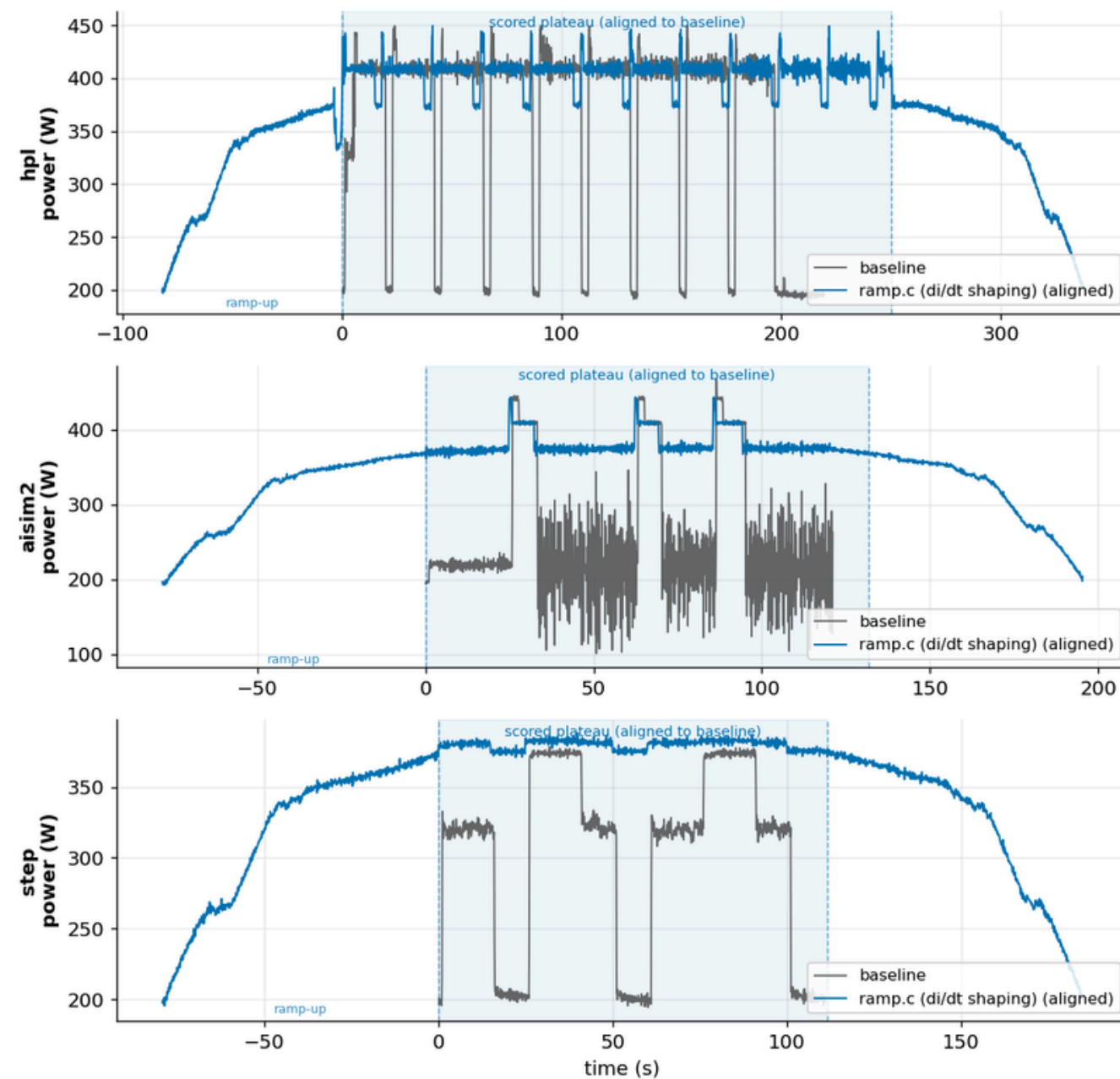
Step Function

- Duration based



RAMP.C RESULTS

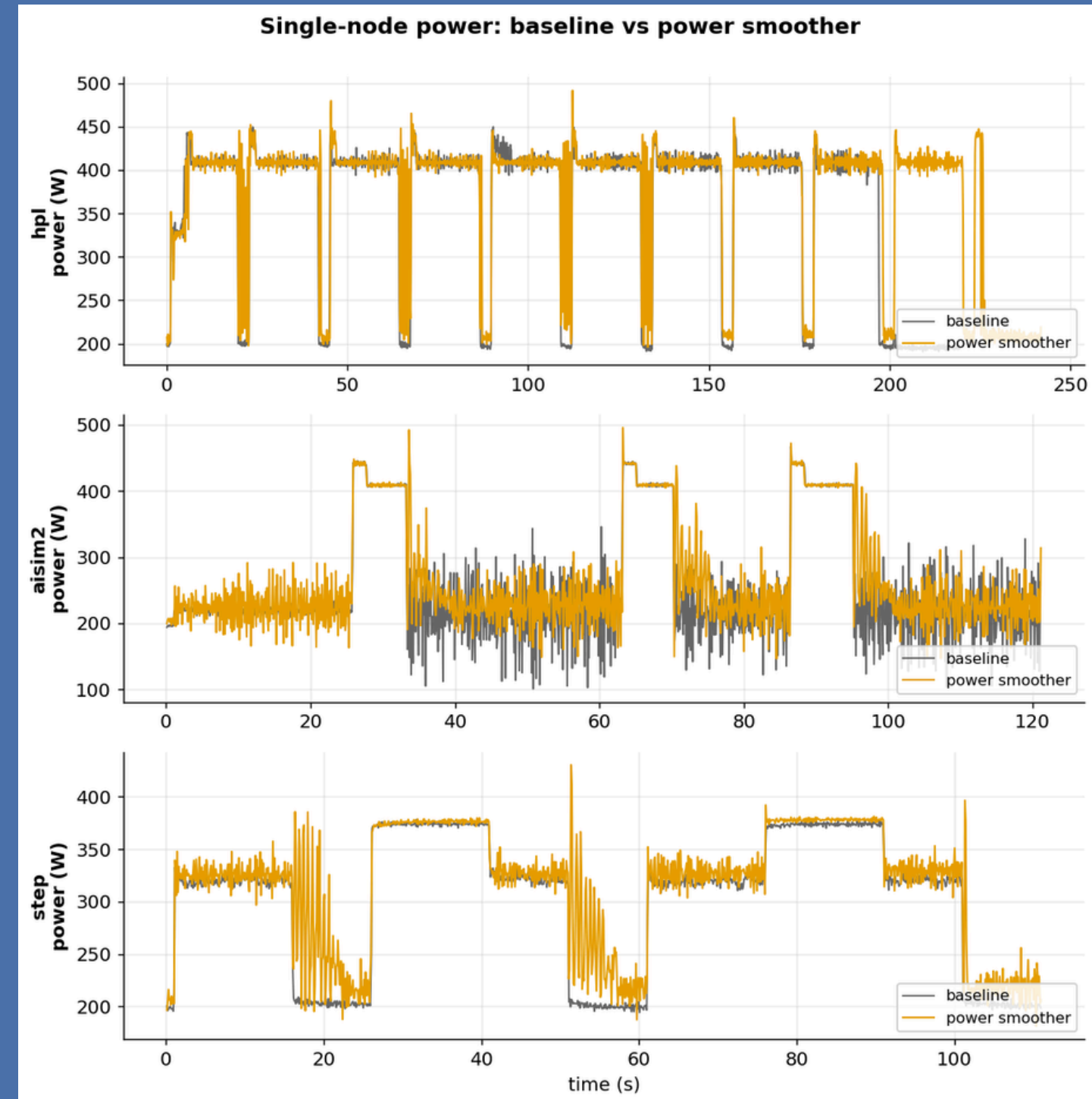
Single-node power: baseline vs ramp.c (di/dt shaping) (untrimmed, plateau aligned to baseline)



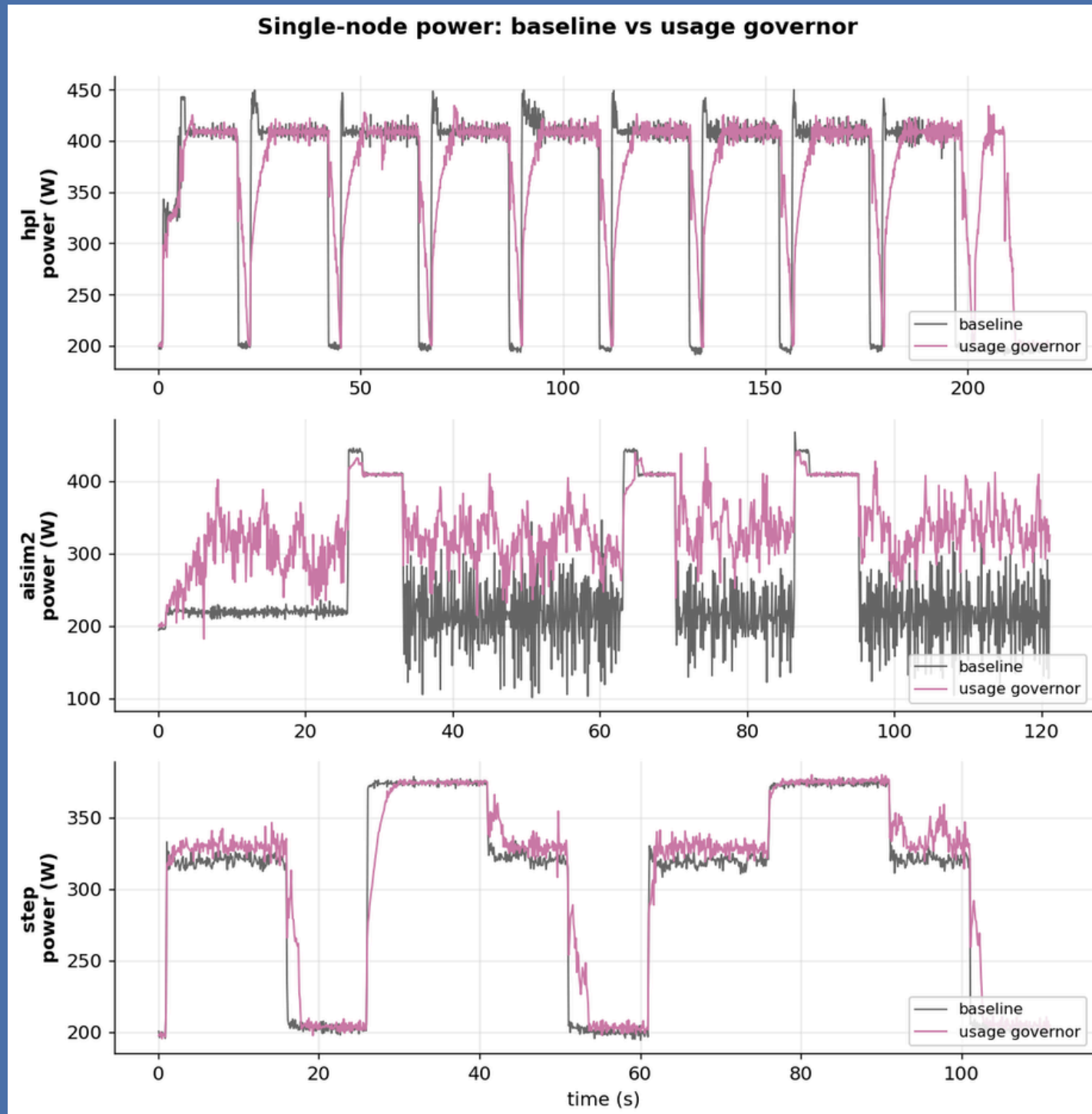
- Very good at smoothing
- Unreasonable cost
 - Time for ramps
 - Energy burnt
 - Reduced GFlops
- Not meant to be deployed

RAMP DOWN RESULTS

- Little impact on drops
 - Doesn't handle spikes
- Lots of noise
- Needs work → deployable



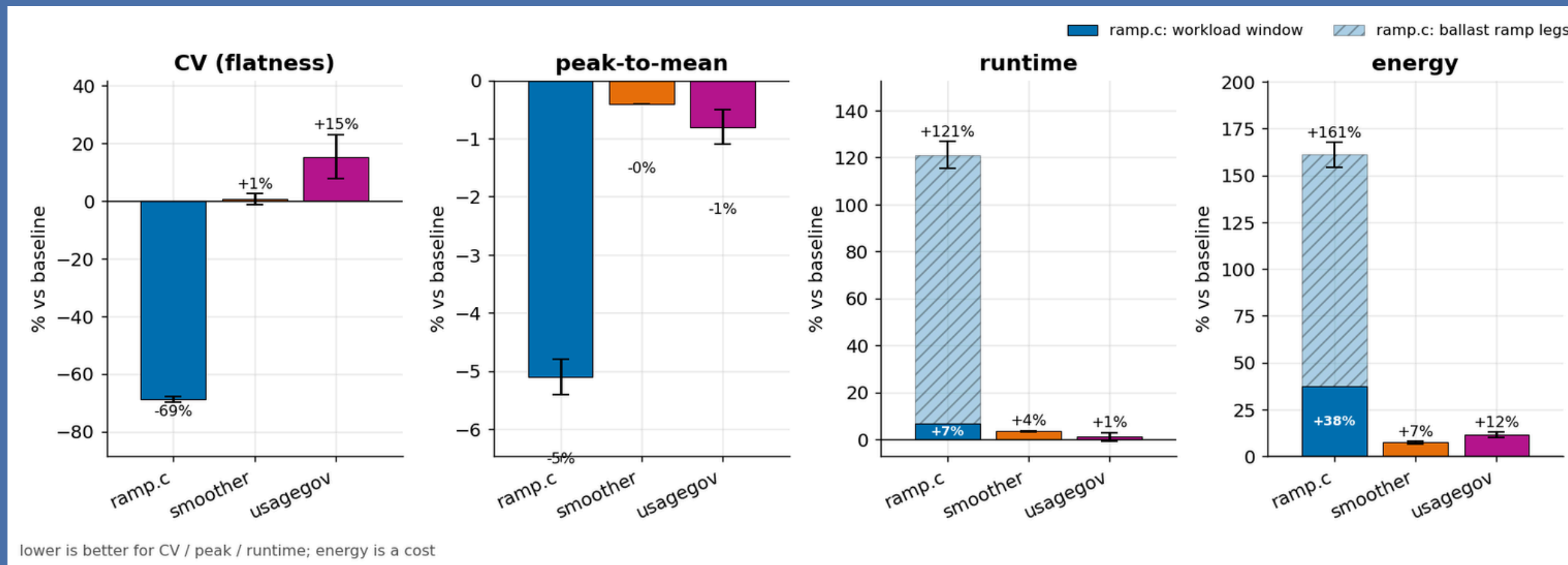
GOVERNOR RESULTS



- Marginally caps power spikes
 - RAPL → reduced GFlops
- Lightweight, but insignificant benefit
- Needs work → deployable

CONCLUSION

- Inconclusive results
 - Limited by time, technical skills
- All 3 take promising approaches
 - Future research → consolidate logic



[1] Choukse, E., Warriar, B., Heath, S., et al. (2025). Power stabilization for AI training datacenters. arXiv. <https://doi.org/10.48550/arxiv.2508.14318>

[2] REDFISH | DMTF. (n.d.). In www.dmtf.org. Retrieved July 24, 2026, from <https://www.dmtf.org/standards/redfish>

[3] Dementiev, T. W. R. (n.d.). Intel® Performance Counter Monitor - A Better Way to Measure CPU... In Intel. Retrieved July 24, 2026, from <https://www.intel.com/content/www/us/en/developer/articles/tool/performance-counter-monitor.html>

[4] Tarara, A. (2023). CPU Utilization - A useful metric? In [Green-coding.io](https://www.green-coding.io). <https://www.green-coding.io/case-studies/cpu-utilization-usefulness/>

[5] North American Electric Reliability Corporation. (n.d.). Reference document: NERC balancing and frequency control. Retrieved July 24, 2026, from https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Reference_Document_NERC_Balancing_and_Frequency_Control.pdf

[6] North American Electric Reliability Corporation. (n.d.). Whitepaper: Characteristics and risks of emerging large loads. Retrieved July 24, 2026, from <https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/whitepaper-characteristics-and-risks-of-emerging-large-loads.pdf>

[7] North American Electric Reliability Corporation. (n.d.). Incident review: Large load loss. Retrieved July 24, 2026, from https://www.nerc.com/globalassets/our-work/reports/event-reports/incident_review_large_load_loss.pdf

[8] North American Electric Reliability Corporation. (2024). 2024 long-term reliability assessment (corrected July 2025). https://www.nerc.com/globalassets/our-work/assessments/2024-ltra_corrected_july_2025.pdf

[9] North American Electric Reliability Corporation. (2025). 2025 state of reliability overview. https://www.nerc.com/globalassets/programs/rapa/pa/nerc_sor_2025_overview.pdf

[10] North American Electric Reliability Corporation. (2025). Effective load carrying capability report. https://www.nerc.com/globalassets/our-work/reports/special-reports/elcc_report_september_2025.pdf

[11] Institute of Electrical and Electronics Engineers. (2018). IEEE Std 1547-2018: IEEE standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces. Retrieved July 24, 2026, from https://en.wikipedia.org/wiki/IEEE_1547

[12] Dominion Energy. (2024). DER interconnection parameters manual. <https://www.dominionenergy.com/-/media/content/large-business-services/pdfs/virginia/der-interconnection-parameters-manual.pdf>

[13] National Renewable Energy Laboratory. (n.d.). Smart inverters applications in power systems (IEEE PES TR67). Retrieved July 24, 2026, from <https://www.nrel.gov/media/docs/libraries/grid/smart-inverters-applications-in-power-systems.pdf>

[14] Best practices for large load interconnections: A North American perspective on data centers. (2025). arXiv. <https://arxiv.org/pdf/2601.12686>

[15] GridLab. (2025). Practical guidance and considerations for large load interconnections: Interim report. <https://gridlab.org/wp-content/uploads/2025/03/GridLab-Report-Large-Loads-Interim-Report.pdf>

[16] Enhancing grid resilience for giga-watt scale data centers using high voltage circuit breaker operated braking resistors. (2024). arXiv. <https://arxiv.org/pdf/2512.21295>

[17] FERC grapples with surging reliability and interconnection demands from data centers. (2025). Orrick. <https://www.orrick.com/en/Insights/2025/11/FERC-Grapples-With-Surging-Reliability-and-Interconnection-Demands-From-Data-Centers>

[18] NERC tees up plan to assess grid risks associated with data centers. (n.d.). White & Case. Retrieved July 24, 2026, from <https://www.whitecase.com/insight-alert/nerc-tees-plan-assess-grid-risks-associated-data-centers>

REFERENCES

[19] How ISOs and RTOs are addressing large load growth in 2025. (n.d.). Yes Energy. Retrieved July 24, 2026, from <https://www.yesenergy.com/blog/how-isos-and-rtos-are-addressing-large-load-growth-in-2025>

[20] Solving interconnection bottlenecks with data center load flexibility. (n.d.). Renewable Energy World. Retrieved July 24, 2026, from <https://www.renewableenergyworld.com/power-grid/solving-interconnection-bottlenecks-with-data-center-load-flexibility/>

[21] Data centers and the grid: How hyperscale computing is reshaping power infrastructure. (n.d.). Power Magazine. Retrieved July 24, 2026, from <https://www.powermag.com/data-centers-and-the-grid-how-hyperscale-computing-is-reshaping-power-infrastructure/>

[22] Resilience assessment and planning in power distribution systems. (2023). arXiv. <https://arxiv.org/pdf/2308.07552>

[23] Department for Energy Security and Net Zero. (2023). Exploring reliability standard metrics in a net zero transition. <https://assets.publishing.service.gov.uk/media/65e3a3323f694514a3035fbe/5-exploring-reliability-standard-metrics-in-net-zero-transition.pdf>

[24] Frequency instability problems in North American interconnections. (2011). U.S. Department of Energy Office of Scientific and Technical Information. <https://www.osti.gov/servlets/purl/1556900>

[25] Kundur, P. (1994). Power system stability and control. McGraw-Hill.

[26] Milano, F., Dörfler, F., Hug, G., Hill, D. J., & Verbič, G. (2018). Foundations and challenges of low-inertia systems. In 2018 Power Systems Computation Conference (PSCC). <https://doi.org/10.23919/PSCC.2018.8450880>

[27] Tielens, P., & Van Hertem, D. (2016). The relevance of inertia in power systems. Renewable and Sustainable Energy Reviews, 55, 999–1009. <https://doi.org/10.1016/j.rser.2015.11.016>

[28] Institute of Electrical and Electronics Engineers. (2019). IEEE Std 1159-2019: IEEE recommended practice for monitoring electric power quality. IEEE.

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Claude code was utilized in the production, operation, and evaluation of software described. Some figures and tables were produced using AI-generated programs.

